

# JUNDA ZHAO

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## Research Interests

Reliable testing and verification for AI coding agents in production: LLM-based test generation from recovered specifications, and the evaluation of AI-generated tests. **Seeking a postdoctoral position starting in late 2027.**

## Education

**Doctor of Philosophy (Ph.D. Candidate)**, Mechanical and Industrial Engineering 09/2022 to present  
**University of Toronto**, St. George Campus, Toronto, Canada

- Advisor: Prof. Eldan Cohen, Optimization and Machine Learning (OptiMaL) Lab; close collaboration with Prof. Shurui Zhou (Electrical and Computer Engineering)
- Dissertation (in progress): *Accurate Specification Recovery for Effective Test Generation*
- Passed the doctoral qualifying examination; expected completion in 2027

**Honours Bachelor of Science**, Computer Science 09/2016 to 11/2021  
**University of Toronto**, St. George Campus, Toronto, Canada

### Selected coursework

- Graduate: CSC2511 Natural Language Computing (A+); MAT1510 Deep Learning: Theory and Data Science (A+); MIE1513 Decision Support Systems (A+)
- Undergraduate: CSC412 Probabilistic Learning and Reasoning (A+/94); CSC413 Neural Networks and Deep Learning (A+/99); CSC420 Introduction to Image Understanding (A+/96)

## Honours and Awards

**ACM SIGSOFT Distinguished Paper Award**, ISSTA 2026 2026  
For “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

**Ontario Graduate Scholarship (OGS)**, Government of Ontario and University of Toronto 2024 to 2025

## Publications

- [1] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study).” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. **ACM SIGSOFT Distinguished Paper Award**. [arXiv](#)
- [2] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests.” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. [arXiv](#)
- [3] **Junda Zhao**, Yuliang Song, Eldan Cohen. “Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models.” *Journal of Systems and Software*, 2025. [Paper](#)
- [4] Faria Khandaker, Arik Senderovich, **Junda Zhao**, Eldan Cohen, Eric Yu, Sebastian Carbajales, Allen Chan. “Transformer Models for Mining Intents and Predicting Activities from Emails in Knowledge-Intensive Processes.” *Engineering Applications of Artificial Intelligence*, 2024. [Paper](#)
- [5] **Junda Zhao**, Weiling Cai, Yizhen Ge, Lijuan Zhang. “A New Fuzzy Belonging-based Multi-view K-means Clustering Algorithm.” *IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON 2019)*. [Paper](#)
- [6] **Junda Zhao**. “Accurate Specification Recovery for Effective Test Generation.” *SPLASH/ISSTA 2026 Doctoral Symposium*.

## Research Experience

**Ph.D. Candidate**, OptiMaL Lab, Department of Mechanical and Industrial Engineering 09/2022 to present  
**University of Toronto**, supervised by Prof. Eldan Cohen, Toronto, Canada

- Reliable testing and verification for AI coding agents, with a focus on LLM-based unit test generation.
- Specification-based testing: recovering the intended behavior of code that may be buggy, and generating tests from it.
- Evaluation of LLM-generated tests and of the metrics used to judge them.
- Code summarization with pre-trained language models of code.
- Public replication packages for published work.

**Research Assistant**, Dynamic Graphics Project (DGP) Lab 11/2021 to 09/2022  
**University of Toronto**, Toronto, Canada

- Research on geometric stylization: developed a method for texture synthesis and transfer based on a feature-grid neural implicit representation of arbitrary objects, whose 2-D structure allows computer vision and 2-D geometry processing techniques to be applied to 3-D representations.

**Undergraduate Projects**, Department of Computer Science, University of Toronto 10/2020 to 09/2021

- **Human musculoskeletal system simulation** (CSC494 research project): proposed, built, and simulated a fundamental unit of the musculoskeletal system from mass points, rods, and springs, with machine learning for fast approximation of the system state.
- **Accurate large-scale ferrofluid simulation** (CSC417 team course project): implemented the ferrofluid simulation of Huang et al. and studied the formation of spikes under a magnetic field.

## Talks and Posters

**Research-track talk and poster**, ISSTA 2026, Oakland, CA, USA 10/2026

“Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

**Research-track talk and poster**, ISSTA 2026, Oakland, CA, USA 10/2026

“Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests”

**Journal-First talk and poster**, ASE 2025, Seoul, South Korea 11/2025

“Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models”

**Doctoral Symposium talk**, SPLASH/ISSTA 2026, Oakland, CA, USA 10/2026

“Accurate Specification Recovery for Effective Test Generation”

## Teaching Experience

**Teaching Assistant, MIE1517 Introduction to Deep Learning** (graduate course) 01/2024 to 04/2024; 09/2024 to present

**University of Toronto**, Department of Mechanical and Industrial Engineering

- Fall and Winter terms since 09/2024. Lead tutorial sessions, help develop and improve course materials and plans, and grade projects and tests.

**Teaching Assistant, MIE370 Introduction to Machine Learning** (undergraduate course) 09/2025 to 12/2025

**University of Toronto**, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

**Teaching Assistant, MIE350 Design and Analysis of Information Systems** (undergraduate course) 09/2022 to 12/2022

**University of Toronto**, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

## Professional Experience

**Junior Identity and Access Management Analyst** 09/2019 to 09/2020

**University of Toronto**, Information Technology Services, Toronto, Canada

- Built and maintained a database system for tracking and analyzing network access and activity information for students, faculty, and staff on the university network.

## Skills

- **Programming:** Python, C++ (Eigen, libigl), MATLAB; Java for test generation and evaluation (JUnit, Defects4J)
- **Machine learning:** PyTorch, NumPy, OpenCV; large language models and deep generative models; statistical analysis and modeling
- **Software testing and analysis:** LLM-based test generation, coverage and mutation analysis (JaCoCo, PIT, CodeCover), experiment design
- **Languages:** fluent in English and Mandarin